

# Does R&D concentration reduce technological diversity?

## Evidence from global top innovators

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### Abstract

This paper investigates whether increasing concentration of R&D expenditure among the world's largest innovators reduces the technological diversity of their innovation output. Using firm-level data from the JRC COR&DIP Top 2000 Innovators matched with patent information from PATSTAT (2010–2022), I construct indicators of sectoral R&D concentration, firm-level technological diversity, and innovation system dynamism. R&D concentration is measured through a Herfindahl–Hirschman Index of sectoral R&D spending, while diversity is captured by a normalized Shannon entropy index of patent portfolios across technological fields. The unbalanced nature of the dataset—where firms enter and exit the global top-2000 ranking over time—is exploited to derive indicators of entry, exit, and rank turbulence as proxies for innovation dynamics. Regression results show that higher sectoral R&D concentration is significantly associated with lower firm-level technological diversity, suggesting that dominant incumbents focus their innovation efforts within narrower technological trajectories. Conversely, larger and more capital-intensive firms tend to display broader technological portfolios, likely exploiting economies of scope. Results on sectoral turbulence are mixed: sectors with higher mobility are associated with higher diversity, but over-time firms might strategically reduce their technological breadth to deter entry. Overall, the findings indicate that growing R&D concentration may undermine technological variety, with implications for innovation and competition policy.

**Keywords:** R&D concentration, technological diversity, superstar firms, innovation policy

**JEL Codes:** O14, Q52, L13

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# 1 Introduction

Over the last two decades, economies and particularly innovation activity have become increasingly concentrated among a handful of large corporations (Bajgar et al., 2023). A growing body of research documents the emergence of “superstar firms” dominating both market shares and R&D spending (De Ridder, 2024). The relationship between concentration and innovation output has been investigated but the debate about either positive or negative monotonic effects is still ongoing.

According to the Schumpeterian hypothesis and the so-called “appropriability issue” (see Levin and Reiss, 1984), concentration (i.e., higher market power) leads to more innovation, since competition reduces expected rents (Romer, 1990; Aghion and Howitt, 1992 and Grossman and Helpman, 1991). On a opposite note, there is a body of empirical literature that found positive effects of competition on innovation (see Blundell et al., 1999; Beneito et al., 2015). Aghion et al. (2005) theorize and test the hypothesis of an inverse-U relationship between competition and innovation which depends on the technology state of different firms<sup>1</sup>. Yet, the implications for the diversity of technological change remain uncovered. This paper seeks to fill this gap by answering the following research question:

*RQ: Does the dominance of a few firms at sector level foster cumulative innovation within narrow technological trajectories, or does it support a broader exploration of new fields?*

To answer this question, I examine whether rising concentration of R&D among the world’s top 2000 innovators reduces the technological diversity of innovation output at the firm-level. Using firm-level data from the JRC COR&DIP dataset matched with patent information from PATSTAT (2010–2022), I construct measures of (i) R&D concentration across firms within each sector, (ii) technological diversity using Shannon entropy of IPC patent classes at firm-level, and (iii) innovation “turbulence,” capturing firm entry, exit, and rank mobility within the global top-innovators list. The empirical strategy employs regressions with country and year fixed effects, linking lagged R&D concentration to subsequent changes in technological diversity.

Preliminary findings suggest that firms operating in more concentrated industries tend to reduce technological diversity. This is consistent with a crowding-out effect of dominant incumbents on technological exploration. At the same time, greater dynamism in the innovation system, measured by the degree of market turbulence, shows evidence of a negative relationship with the rate of technological diversification. This result could arise from the strategic behavior of incumbent firms, which find themselves narrowing the technological

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<sup>1</sup>These authors were assigned the Nobel Prize in Economics in October 2025.

scope of their innovation efforts to ensure a competitive advantage when the degree of turbulence is high.

These findings align with the most recent theory (Aghion et al., 2005) according to which there is no monotonic relationship between concentration and innovative output and speak directly to ongoing policy debates on competition and innovation policy in advanced economies. They quantify the trade-off between industrial concentration and technological variety, which is a major concern for the innovation and industrial-policy agenda.

## 2 Related literature

### 2.1 Competition and Innovation

The relationship between market competition and innovation remains one of the central debates in industrial organization and innovation economics. Following the classical Schumpeterian hypothesis, higher market concentration - i.e., greater monopoly power — is thought to foster innovation by increasing firms’ expected rents and thus their incentives to invest in R&D (Romer, 1990; Aghion and Howitt, 1992; Levin and Reiss, 1984). The underlying idea is that competition erodes profit margins and therefore reduces firms’ ability and willingness to undertake risky innovative investments. Early empirical contributions (e.g. Levin and Reiss, 1984; Arrow, 1972) support this view, showing that market power can enhance firms’ capacity to appropriate the returns from innovation.

However, more recent theoretical and empirical work challenges this monotonic relationship. Aghion et al. (2005) introduced the “inverted-U” hypothesis, demonstrating that the effect of competition on innovation depends on firms’ technological distance. In industries where firms are close to the technological frontier, competition spurs innovation as firms race to escape rivals; in contrast, for laggard firms far from the frontier, competition discourages R&D since the probability of catching up is small. Blundell et al. (1999) and Beneito et al. (2015) provide supporting evidence at the firm level, showing that moderate competition pressures can indeed stimulate innovation output.

This nuanced relationship between competition and innovation highlights the importance of considering market structure and concentration jointly with innovation dynamics. Vives (2008) further emphasizes that competition should not be measured only by the number of firms in a market, but also by the intensity of their strategic interactions and the degree of market integration. In sectors characterized by strong complementarities and spillovers, collaboration and openness may enhance, rather than limit, innovative performance. The notion of clusters developed by Porter (1991) and Porter and Linde (1995) illustrates this

mechanism: geographically or technologically proximate firms benefit from localized knowledge exchange, shared resources, and cumulative learning effects, which translate into higher innovation output (Feldman, 1999; Oerlemans et al., 2001).

In this context, market concentration does not merely capture competitive pressure — it also shapes the environment in which firms can learn, recombine technologies, and reallocate resources toward innovation. A high level of concentration may reflect efficiency and scale, but can also reduce entry and dynamism, limiting the renewal of leading innovators. Conversely, more competitive and diversified environments can sustain innovation through the continuous reshuffling of technological leaders. This perspective is particularly relevant for the global R&D frontier, where large incumbent firms dominate innovation investment, but where new entrants and technological diversification can still generate substantial creative renewal.

## **2.2 Diversity, concentration and turbulence**

### **2.2.1 Technological diversity**

Technological diversity measures the variety of technological classes in which the innovating firm concentrates its activity over time. This characteristic has been studied in the literature from several perspectives, starting from its determinants. In particular, against a basic hypothesis of randomness in the choice to increase the technological scope of innovations, Breschi et al. (2003) find significant evidence of a strategy based on knowledge relatedness. In other words, firms constantly learn and progressively fill the gaps by extending their technological scope to adjacent fields. Furthermore, in the same study, this effect is found to be correlated with size.

Technological diversity has been investigated in its effects on business and innovation performance. Gambardella and Torrisi (1998) find that it positively impacts indicator of business performance as sales, profits and sales intensity (i.e., scaled by size). These results are interpreted by the authors as indicative of the fact that innovating on multiple fronts is not only a way to access new markets but can also improve the quality of existing products. As far as innovation performance is concerned, Garcia-Vega (2006) find that diversity enhances output, thanks to internal spillovers and reduced investment risk. On a similar note, Bolli et al. (2020) find a positive impact in the discovery stage, allowing for spillovers and knowledge recombination (Moaniba et al., 2018)<sup>2</sup>, but a negative effect in the commercialization stage where it is inversely correlated with profits.

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<sup>2</sup>While these factors are positive for fostering relevant innovation, the impact on other dimensions of firms' strategies can be negative: Marhold and Kang (2017) find evidence of a negative relationship between firm-level technological diversity and its portfolio of alliances, in a exploration-exploitation paradigm.

Patent databases generally provide a range of useful information for classifying inventions. These include IPC/CPC codes or technology classes. Many papers use this information to construct technological diversity indices at the firm level (see seminal contributions by Lerner, 1994 and Hall et al., 2001). Besides simple indicators (e.g., ratios or technology counts) which are usually used as alternative or robustness measures, articles dealing with technological diversity mainly choose between concentration indices and entropy measures. Hsu et al. (2018) exploit the technology classification provided by the NBER patent database (Hall et al., 2001) and measure diversity as one minus the technological concentration (compiled with an Herfindahl Index) on innovations patented in the last three years for each firm. Huang and Chen (2010) employ an entropy measure based on firm-level technological differentiation (4-digits IPC codes).

Concentration indices, originating from the literature on industrial organization, focus on the dominance of certain categories (or firms) over others. In contrast, entropy indices, originating from information theory<sup>3</sup>, measure dispersion within a given distribution. This difference allows the latter to capture not only the balance but also the richness of categories (Miller, 2006; Frenken et al., 2007; Nesta and Saviotti, 2005). In other words, where the concentration index measures the dominance of one or more categories over others, entropy measures are more suitable to consider the breadth of the technological portfolio in the perspective that greater diversity means greater uncertainty with respect to the outcome of the random selection of a technology. This added complexity of the Shannon Entropy Index lead me to prioritize it as a technology diversification measure.

### **2.2.2 Concentration**

In line with the industrial organization literature, I use the Herfindahl–Hirschman Index (HHI) as a measure of market concentration, in this case applied to the R&D expenditure for top global innovators.

### **2.2.3 Turbulence**

Similarly to technological diversity, market turbulence has also been addressed in the literature both from the perspective of dominance - or, in other words, concentration persistence - and dispersion in the distribution of companies on the market. Mataves and Rondi (2007) measure turbulence as one minus the ratio between the cumulated market share of the first 5 firms at the beginning of the panel period over the concentration of the first 5 firms at the end of the period. The second approach to measuring market turbulence considers firm entry,

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<sup>3</sup>see the seminal contribution by Shannon, 1948 and Jacquemin and Berry, 1979

exit, and internal market dynamics<sup>4</sup>. These measures arise from the literature on business dynamics, i.e., studying the entry-exit behavior of firms (turnover) as well as variations in market shares of incumbents (mobility), with both empirical and theoretical contributions (Caves, 1998).

This approach, used in this paper, aims to consider both the impact of firm turnover and the stability of incumbents in the market (in the fashion of Dosi et al., 1995). Taking advantage of the characteristic composition of this dataset, this latter variable is expressed in the stability of the ranking over the observation period.

### 3 Data

The empirical analysis draws on the *JRC COR&DIP* database of the top 2000 global R&D investors<sup>5</sup>, covering the period 2010–2022. The dataset provides harmonized information on firms’ R&D investment, capital expenditures, net sales and operating profits (million \$), number of employees. These firm-level data are matched to patent applications ( $N_{pat}$ ) from the *PATSTAT* database, aggregated by priority year and International Patent Classification (IPC) classes. After matching, the resulting unbalanced panel includes 3,946 firms across 17 NACE 2-digit sectors<sup>6</sup> (see Table 1 for descriptives).

| Variable  | N      | Mean     | SD       | Min       | p25      | p50      | p75      | Max      |
|-----------|--------|----------|----------|-----------|----------|----------|----------|----------|
| R&D       | 27,853 | 361.4043 | 1141.163 | .020402   | 50       | 92.66569 | 221      | 37033.58 |
| Net Sales | 27,511 | 9103.131 | 24467.19 | 0         | 675.1653 | 2174.482 | 7010.465 | 601215.1 |
| R&D.int_w | 27,407 | 1.67354  | 14.92484 | .0002615  | .0248131 | .0518754 | .1370161 | 520.3113 |
| Capex     | 26,305 | 624.2138 | 2300.411 | 0         | 22.6     | 87.39874 | 340.9196 | 78460.1  |
| Op_Profit | 27,814 | 941.6825 | 4606.832 | -27703.38 | 25.12891 | 165.9949 | 595.3462 | 286755.5 |
| Emp       | 25,707 | 27201.83 | 55473.75 | 0         | 2600     | 8800     | 24827    | 961000   |
| N_pat     | 19,803 | 269.1972 | 912.5889 | 1         | 11       | 42       | 162      | 25399    |

Table 1: Descriptive statistics

Statistics in Table 1 highlight the strong heterogeneity in firm size and R&D intensity, with mean R&D spending around \$361 million and median employment of about 8,800 employees. The distribution of both R&D expenditures and patents is highly skewed, reflecting the well-known concentration of innovation activity among a small number of global leaders. These descriptive patterns are consistent with prior evidence on “superstar” R&D performers (see Aghion and Howitt, 2022; De Ridder, 2024). R&D intensity is measured as the ratio of

<sup>4</sup>see Davis and Haltiwanger, 1992 for a turnover measure in a job reallocation and firm dynamics study.

<sup>5</sup>See more at <https://data.jrc.ec.europa.eu/collection/id-00156>

<sup>6</sup>See <https://ec.europa.eu/eurostat/documents/3859598/5902521/KS-RA-07-015-EN.PDF.pdf/dd5443f5-b886-40e4-920d-9df03590ff91?t=1414781457000>.

R&D and Net Sales, which were winsorized at 0.01 level to exclude some outliers<sup>7</sup> and have more reliable ratios without altering distributions. To these variables I add a measure of sectoral-level concentration and a measure of firm-level technological diversity. Furthermore, since this dataset is not a proper panel, I insert a few measures to proxy dynamism in the dataset. In particular, I add two firm-level variables: a dummy indicating whether the firm entered the panel for the first time in a given year and a continuous variable indicating the average variability of a firm global ranking in the observation period. Then I combine and weight this information to create an index of market share turbulence.

R&D concentration is measured using the Herfindahl–Hirschman Index (HHI):

$$HHI_{s,t} = \sum_{i \in s} \left( \frac{RD_{i,s,t}}{\sum_{i \in s} RD_{i,s,t}} \right)^2 \quad (1)$$

where a higher value indicates greater concentration of R&D expenditures among a few firms within sector  $s$  and year  $t$ . For the sake of simplicity, the HHI, which usually is between 0 and 10,000, is rescaled from 0 to 1. While the average yearly HHI has been increasing from 0.026 in 2009 to 0.0346 in 2022 (31%), it remains overall quite low. Looking at specific sectors, the most representative (Manufacturing) increased of only 6%, while others increased way more: wholesale and retail trade (the code referred to Amazon, among others) has increased by more than 37%; information and communication increased by more than 66%.

Technological diversity is captured using the Shannon entropy of the distribution of a firm’s patent portfolio across technological fields. For each firm  $i$  in year  $t$ , let  $p_{k,i,t}$  denote the share of patents belonging to technology class  $k$ . The (unnormalized) entropy is given by

$$H_{i,t} = - \sum_k p_{k,i,t} \ln(p_{k,i,t}),$$

where a higher value indicates a more even distribution of patents across classes. To facilitate comparison across firms, I normalize this measure by the logarithm of the total number of distinct technology classes that appear in the global dataset in each year:

$$H_{i,t}^{\text{norm}} = \frac{H_{i,t}}{\ln K_t},$$

where  $K_t$  is the number of unique WIPO (or IPC) technology classes in which any firm patents during year  $t$ . This normalization ensures that firms are evaluated relative to the technological space available in that period, so that, for instance, a firm active in three

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<sup>7</sup>although negative values are technically possible, their incidence in variables such as R&D, Capex or Net Sales is low enough ( $\leq 1\%$ ) to consider them as outliers. Nevertheless, results of the empirical analysis are robust to the non-winsorized R&D intensity variable.



technology fields out of one hundred is considered less diversified than one active in the same number of fields when only ten were available.

Moreover, the COR&DIP Top Innovators dataset is an unbalanced panel, as firms appear intermittently across survey waves depending on their inclusion among the top 2000 R&D performers in each year. This feature reflects the dynamic composition of the global innovation frontier: new firms enter the ranking as their R&D investments rise, while others drop out as their relative effort declines. The analysis exploits it to construct indicators that can control for firm-level and overall innovation system dynamism.

In particular, entry (*Entry*) and exit (*Exit*) are dummy variables equal to 1 if the firm  $i$  entered or exited the sample in year  $t$ , while the within-firm variance of global rank (*Rank\_var*) measures how much the relative position of incumbents fluctuates over time. To these measures I add an index of market share turbulence. I first compute a sectoral turnover rate based on firm entry and exit in the Top 2000 list:

$$Turnover_{s,t} = \frac{\sum_i Entry_{i,s,t} + \sum_i Exit_{i,s,t}}{(N_{s,t} + N_{s,t-1})/2} \quad (2)$$

where  $\sum_i Entry_{i,s,t}$  and  $\sum_i Exit_{i,s,t}$  are the counts of entering and exiting firms in year  $t$ , and  $N_{s,t}$  is the total number of firms in sector  $s$  at time  $t$ . This measure captures the renewal of leading innovators. Then, I use this Turnover indicator to compute a standardized turbulence index at firm-level (*Turb\_z*), incorporating the change in rank that an incumbent firm experiences from one year to the following one, as:

$$Turbulence_{i,t} = w \cdot z(Turnover_{s,t}) + (1 - w) \cdot z(\Delta Rank_{i,t}) \quad (3)$$

where  $w$  is a weight set at 0.5. These indicators make it possible to assess how the structure of corporate R&D evolves even in the absence of a balanced firm-level panel, providing a way to quantify the intensity of competitive turnover among the world's main innovators.

## 4 Empirical strategy and results

The baseline specification relates technological diversity to R&D concentration and sector- or firm-level controls:

$$Diversity_{it} = \beta Concentration_{st} + \gamma Controls_{it} + \alpha_s + \lambda_t + \epsilon_{it} \quad (4)$$

All models include year and country fixed effects; standard errors are clustered at the

firm level. In extended specifications, I add sectoral entry and turbulence measures to test whether market dynamism mediates the effect of concentration, and sector fixed effects. To have more interpretable coefficients, capex and employees variables were transformed in logs.

| Diversity    | (1)                     | (2)                     | (3)                     | (4)                     |
|--------------|-------------------------|-------------------------|-------------------------|-------------------------|
| HHI_sec      | -0.189***<br>(0.000)    | -0.180***<br>(0.000)    | -0.257***<br>(0.000)    | 0.0580<br>(0.597)       |
| R&D_int      | -0.000232***<br>(0.000) | -0.000222***<br>(0.000) | -0.000219***<br>(0.001) | -0.000240***<br>(0.000) |
| ln_Capex     | 0.00955***<br>(0.000)   | 0.00974***<br>(0.000)   | 0.0102***<br>(0.000)    | 0.0108***<br>(0.000)    |
| ln_Emp       | 0.0316***<br>(0.000)    | 0.0304***<br>(0.000)    | 0.0307***<br>(0.000)    | 0.0308***<br>(0.000)    |
| Entry        | -0.0205***<br>(0.000)   | -0.0220***<br>(0.000)   |                         |                         |
| Rank_var     |                         | -9.59e-08***<br>(0.000) | -8.25e-08***<br>(0.000) | -7.07e-08***<br>(0.001) |
| Turb_z       |                         |                         | 0.0120***<br>(0.000)    | -0.0198***<br>(0.000)   |
| Observations | 13,757                  | 13,752                  | 13,727                  | 13,726                  |
| R-squared    | 0.242                   | 0.242                   | 0.246                   | 0.257                   |
| Year FE      | Yes                     | Yes                     | Yes                     | Yes                     |
| Sector FE    | No                      | No                      | No                      | Yes                     |
| Country FE   | Yes                     | Yes                     | Yes                     | Yes                     |
| r2_a         | 0.238                   | 0.239                   | 0.242                   | 0.253                   |

Robust pval in parentheses  
\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 2: Regression analysis. Dependent variable: firm-level technological diversity index

Results from Table 2 confirm that concentration within sectors is associated with reduced within-firm technological breadth. The same goes for firms with high R&D intensity level, suggesting that increasing research effort tends to be constrained on a tighter set of technologies. Higher capital expenditure is robustly associated with higher diversity, possibly reflecting specialization in capital-intensive technologies (as is the case for automotive or semiconductors). At the same time, the positive effect of employment shows that larger firms diversify across more technological domains, consistent with economies of scope in R&D. Firm-level entry dummy shows a negative coefficient, as is the case for the rank variation. This means that new entrants generally focus on less technologies in order to emerge

in the market while rank variability may be a proxy for instability, which in turn leads to the reduction of technological diversity. In fact, even if rank variability is directly associated with variation in R&D investment, this one is tied to company performance, and is among the first expenditure items to be affected by a crisis. The coefficient for sectoral turbulence in Column (3) is positive and significant, showing that higher mobility of firms as well as ranking variability at sector level is associated with higher diversity.

In column (4) I introduce sector fixed effects, shifting the interpretation of sectoral variables to a within-sector change over time. Concentration index becomes non-significant, showing that the effect of concentration is completely captured by sector fixed effects. On a different note, the negative coefficient for market share turbulence suggests that in case of higher dynamism, incumbent firms tend to narrow the scope of their innovation effort over time within the same sector, probably as a strategic response to new entrants' competition, in line with results from Toh and Kim (2013).

While this might seem in contrast with the result on sectoral concentration, the interpretation of these results is not trivial. In fact, on the one hand, it is important to note how dynamism variables are firm-level proxies for instability of firm ranking, which might be consequence of several factors (e.g., lower (higher) R&D investment due to performance below (above) expectations). These factors could lead firms to increase (reduce) the technological breadth of their innovation effort according to their good (bad) performance. On the other hand, the relationship between competition and innovation was found to have an inverted-U shape (Aghion et al., 2005) suggesting that both competition and concentration may be coherently associated with reduced innovation. With this paper, I find evidence that this might be the case for technological diversity as well.

## 5 Conclusion

The findings suggest that increasing R&D concentration — especially in ICT and pharmaceutical sectors — may hinder technological variety, potentially slowing cross-domain innovation and diffusion. At the same time, maintaining a dynamic environment with frequent entry and high mobility among innovators could lead to a similar outcome resulting from strategic behavior.

Future developments of this work include the introduction of nonlinearities and robustness analyses to deepen and strengthen the results. The data section will be improved with some more descriptive statistics and graphs.

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# Appendices